



EUROPEAN **Mi**CROSOFT **FABRiC**

Community Conference

STOCKHOLM 24-27 SEPTEMBER 2024

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#FABCONEUROPE



The Data Dojo

A Power BI
Community of Practice

James D. Bartlett III

DataVolume.xyz

@JamesDBartlett3



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About the Speaker

Intro

- James D. Bartlett III (@JamesDBartlett3) {he/him/his}
- Senior BI Analyst at Des Moines University in Iowa, USA

Professional Credentials

- Microsoft MVP (Most Valuable Professional)
- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)
- Microsoft Certified: Power BI Data Analyst Associate (PL-300)

Extracurricular Activities

- User Groups
 - Iowa Power BI User Group (Co-Organizer)
 - Higher Education Power BI User Group (Co-Organizer)
- Open-Source Projects
 - Fabric-Archive-Bot
 - ps-for-pbi
 - PowerBits
 - PoshBits

Opinions expressed are solely my own, and do not represent the views of my employer, nor of any other organization with which I am affiliated.



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Session Agenda

Founding the Data Dojo

Recruiting Data Dojo Members

Introductory Workshops

Our First Major Setback

The Data Dojo Strikes Back!

Hands-On Workshops

A New Event Format Appears!

Resources

Q&A



Founding the Data Dojo

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What is a Community of Practice?

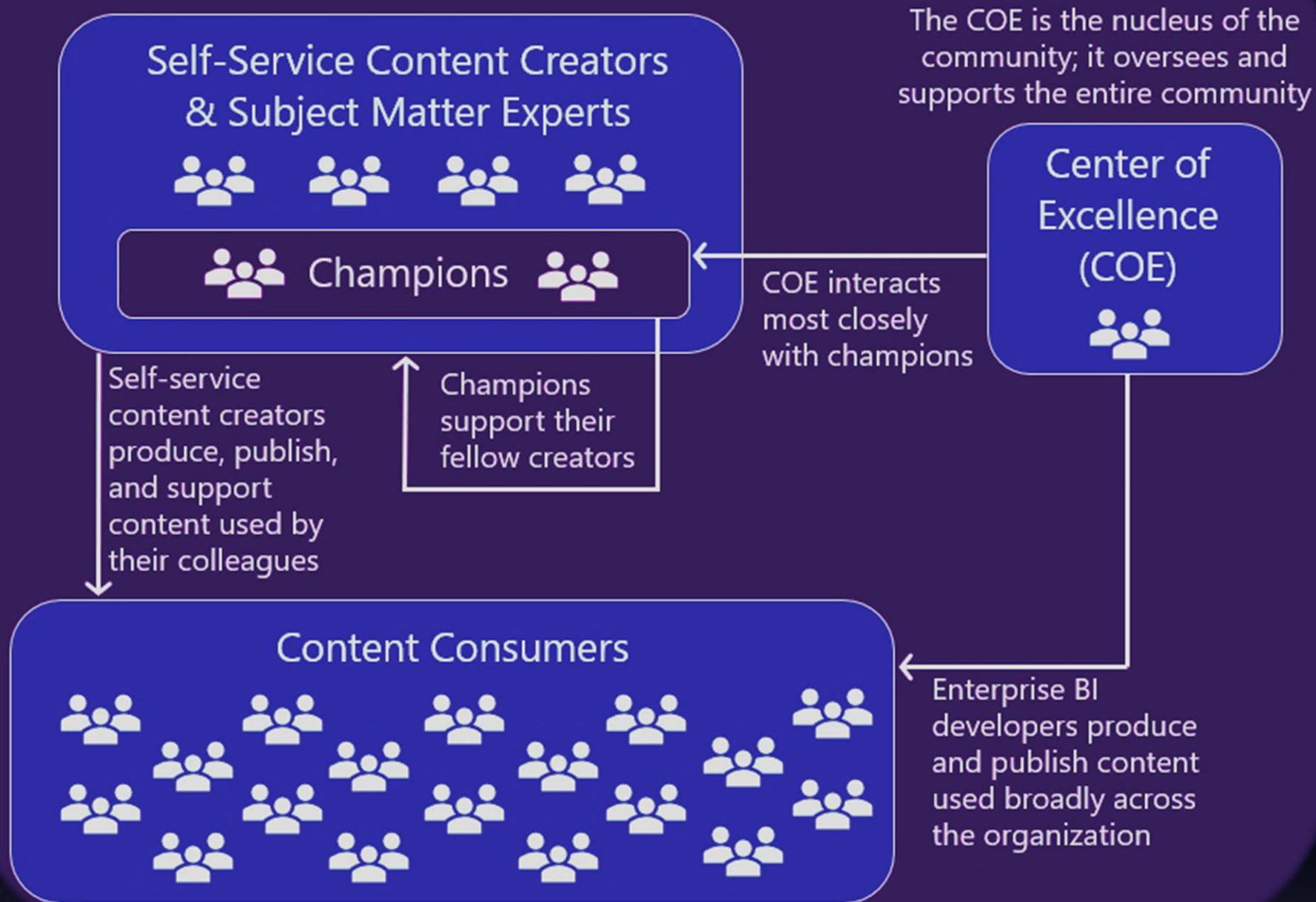
“Communities of Practice are groups of people who share a concern or a passion for something they do, and learn how to do it better as they interact regularly.”

– Etienne and Beverly Wenger-Trayner, June 2015



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/* ----- Note ----- */  
'Community of Practice' <> 'Center of Excellence'
```

Power BI Community of Practice



Data Dojo Backstory: The Status Quo

- Des Moines University (DMU) is a graduate-level medical school
- Most faculty & staff have advanced degrees in STEM
- Strong culture of evidence-based, data-driven decision making
- Not-for-profit org – budget is often a limiting factor
- Huge backlog of data projects waiting for time and/or funding

Data Dojo Backstory: A New Paradigm

- Decentralized, democratized data analytics development
 - a.k.a. “Self-Service Business Intelligence”
- Inter-disciplinary knowledge sharing
- Intra-organizational communication
- Cross-functional, silo-busting collaboration

Founding Principles of the Data Dojo

1. We welcome anyone to join – no gatekeeping allowed!
2. Our events and activities are 100% optional
3. Our members are encouraged (but never required) to share knowledge and collaborate
4. We facilitate collaboration and knowledge sharing via:
 - Interactive workshops and other events
 - Our private Data Dojo group on Microsoft Viva Engage

Founding Principles of the Data Dojo

- 5. We have no instructors or students – we all learn from each other
- 6. We believe the only “stupid question” is the one that goes unasked
- 7. We vote on dates and topics for upcoming events
- 8. We encourage our members to network and build intra-organizational relationships

Calling All Data Ninjas



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Recruiting Data Ninjas (Data Dojo Members)

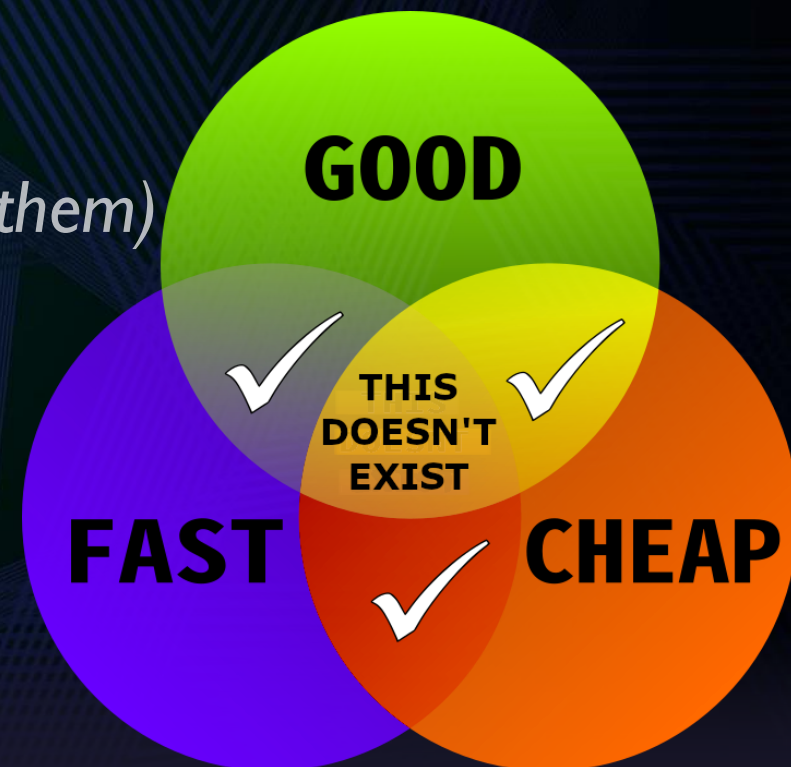
1. Create a Microsoft Forms survey to gauge respondents' interest, availability, current skill level, etc.
2. Create a private Data Dojo group in Viva Engage
3. Utilize the PBI Monitor report (Power BI activity logs)
 - Find the top 20% of Power BI Pro licensed users by activity count over the past 3 months
4. Send Viva Engage group invitation and MS Forms survey link
5. Repeat steps 3-4 quarterly

Introductory Workshops



Welcome to the Data Dojo

- Introductions around the room
- What we're doing here, and why it matters
- Data growth – size, sources, solutions, significance, and silos
- The importance of speed, quality, and cost in data analysis (*and finding the right balance between them*)



5 Types of Data Analytics

1. Descriptive – “What happened, where, and when?”
2. Diagnostic – “Why did it happen?”
3. Predictive – “What will happen in the future?”
4. Prescriptive – “What should we do about it?”
5. Cognitive – “How can we do it better?”

Forms and Formats of Data

- Clean vs. Dirty Data

- Ex: Enterprise Data Warehouse vs. Network folder full of Excel and JSON files

- Structured vs. Unstructured Data

- Ex: Relational Database (SQL Server) vs. Non-Relational Database (NoSQL)

- Quantitative vs. Qualitative Data

- Ex: Average Student Exam Score vs. Student Course Evaluation Sentiment Analysis

The Data Analytics Cycle



Status Quo, Future Vision, Testimonials

- The current paradigm and present state of data analysis
- Our vision for the future and how we plan to make it happen
- Testimonials from Data Ninjas with prior Power BI experience
- “Better Instruction Through Data” (a member presentation)
 1. Live demo of reports made using curated data infrastructure provided by IT department
 2. Stories of how the insights from these reports are making an impact in the business
 3. Challenges encountered and overcome through knowledge sharing and collaboration

Our First Major Setback



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Pitfalls on the Path

My 3 Big Mistakes as a Data Dojo Organizer

1. Poor Planning and Communication
2. Inconsistent Engagement
3. “Letting Perfect be the Enemy of Good”

Mistake #1: Poor Planning & Communication

- Plenty of momentum, but no actual roadmap
- Tried to find dates that would work for all Data Ninjas
- Kept postponing the next event indefinitely
- Failed to communicate with the Data Ninjas about these delays

Mistake #2: Inconsistent Engagement

- Failed to continue recruiting new Data Ninjas
- Failed to consistently engage with existing Data Ninjas
 - Wasn't proactive enough in seeking feedback and suggestions for future topics
- Failed to fuel group discussions with new posts, polls, articles, etc.
 - Most interactions were between the same handful of people
 - Lively conversations were few and far between

Mistake #3: Letting Perfect be the Enemy of Good

- Had a “grand vision” for the next event
- Failed to anticipate complications and delays
- Allowed perfectionism to obstruct progress

“No plan of operations can with any certainty reach beyond the first encounter with the enemy.”

- Helmuth von Moltke, 1871

The Data Dojo Strikes Back!



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The Art of War: Strategy First, Then Tactics

- Zoom out, re-calibrate, and then zoom back in
- Target attainable goals first, then set “stretch goals”
- “Beware of tunnel vision masquerading as laser focus”

“Strategy without tactics is the slowest route to victory.
Tactics without strategy is the noise before defeat.”

– Sun Tzu, “The Art of War”



Wake the Sleeping Dragon

- Begin scheduling events ASAP
- Engage and communicate with the Data Dojo on a regular basis
- Recruit co-organizers from within the ranks of Data Ninjas
- Encourage Data Ninjas to “show and tell” their projects
- Experiment with new event formats, and refine the existing formats.



Hands-On Workshops



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Power BI Desktop & Data Dojo Report Template

- Installing Power BI Desktop from the Microsoft Store
- Opening and Touring the Data Dojo Report Template
 - Dummy financial data (flat file, included with Power BI Desktop)
 - How to normalize (split into dimension and fact tables) using Power Query
 - How to create a Star Schema in Model View
 - Measures
 - Implicit vs. Explicit
 - How to write a simple explicit measure in DAX
 - Filters
 - Filter Pane Sections: Report-Level, Page-Level, and Visual-Level
 - Report-Level (Filter Pane) vs. Model-Level (Power Query)

Data Dojo Report Template – Advanced Features

- Utility Tables
 - __DataSource_Info
 - __SemanticModel_Info
- Custom Power Query (M) Functions
 - fn_GetTableFromDataflow
 - fn_GetLocalTime
 - fn_Coalesce
- Power Query (M) Parameters
 - Source_Workspace
 - Source_Dataflow
 - Row_Limit
- Slicers w/ dynamic (DAX) background color indicating selection state
- Data Freshness Indicator (DFI) – dynamic icon on report page w/ details tooltip

The Importance of Hands-On Workshops

1. Slideshows are *boring*

- Boring to make
- Boring to watch – *except **this** one, of course!* 😄

2. Technical concepts and skills are best learned...

- Via live demonstration
- In an interactive, hands-on format

3. Following along with a live demo is...

- More memorable (and thus more effective as a learning aid)
- More fun (and thus more likely to draw greater attendance)

A New Data Dojo Event Format Appears!



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Traditional Power BI Workshops

- Hosting a traditional Power BI workshop requires
 - Time
 - Energy
 - Patience
 - Coordination
 - (none of which are in plentiful supply)
- Keeping a consistent schedule of workshops is difficult at best
- But our Data Ninjas need to stay engaged and continue learning
- What's a Data Dojo organizer to do?

New Format: Office Hours & Coffee Lounge

Inspired by two scenes often found on a university campus

1. Office Hours

- Consistent schedule of days and times
- Data Ninjas can visit at any time during those hours
- No specific topics, just answering any questions that come up
- Organizer works on their usual tasks until someone shows up (no “lost” time)

2. Coffee Lounge

- A place to gather, hang out, and have discussions
- A casual environment with no expectations
- Socializing is encouraged (it helps build and strengthen relationships)

Data Dojo Resources



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Data Dojo Resources



Q & A





Thank You!

Please rate my session
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James D. Bartlett III

DataVolume.xyz

@JamesDBartlett3 on [GitHub](#), [LinkedIn](#), [Bluesky](#), and [Mastodon](#)

